

THE UNITED REPUBLIC OF TANZANIA
PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
DARES SALAAM REGION
FORM FOUR - MOCK EXAMINATION - 2026
PHYSICS 2A (ACTUAL PRACTICAL)

CODE:031/2A
TIME:2:30Hours

Wednesday 20th May, 2026 A.M

INSTRUCTIONS

1. This paper consists of two (2) questions.
2. Answer all questions.
3. Each question carries twenty **five (25)** marks.
4. Non-programmable calculators may be used.
5. Mobile phones, smart watches and any other unauthorized gadgets are **not** allowed in the examination room.

gepam science hub.com

This paper consists of 3 printed pages

1. The aim of the experiment was to determine the density of a meter rule
Proceed as follows:
 - (a) Locate the centre of gravity C of the meter rule EF by balancing it freely on the triangular prism/knife edge.
 - (b) Suspend 100 g mass on the ruler with distance $d = 10$ cm from C, adjust the position of the prism to get a balance as shown in the following figure.

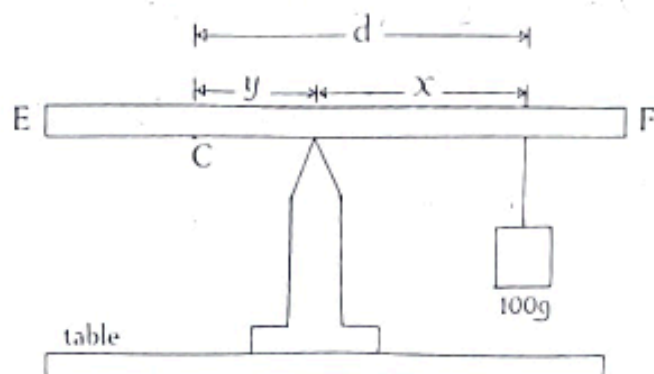


figure 1

- (c) Record the distance y from the centre of gravity to the prism and distance x from prism to the known mass of 100 g
- (d) Repeat the procedures in 1(b) and (c) by increasing the distance of 100 g to $d = 15$ cm, 20 cm, 25 and 30 cm

Questions

- (i) Tabulate the results in a suitable table showing the values of d , x and y .
 - (ii) Plot a graph of y (cm) against x (cm).
 - (iii) Determine the slope of the graph.
 - (iv) Measure and record the length, width and the thickness of the meter rule provided.
 - (v) Determine the density of the meter rule and mass of a meter rule.
 - (vi) Describe how the slope obtained from the graph is related to the mass of the meter rule. **(25 marks)**
2. The aim of this experiment is to determine the refractive index n of a given glass block

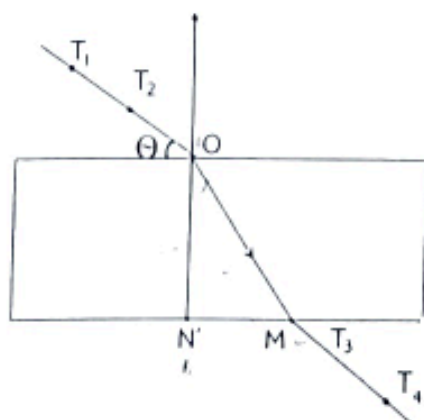


figure 2.

- (a) Place the glass block on the white paper on a drawing board. Using a pencil, trace the outline of the block.
- (b) Remove the glass block and draw the normal line NO near the left end of the block.
- (c) Using a protractor and pencil measure the angle 20° with the surface RR of the block, erect two pins T_1 and T_2 as shown above.
- (d) Replace the glass block to its holder sheet of paper.
- (e) Stick the pins T_3 and T_4 at a position such that they are in straight line with pins T_1 and T_2 as seen through the block. Remove the block and draw a complete path of the ray.
- (f) Measure the length MN' and ON' .
- (g) Repeat the procedures for the values of $\theta = 30^\circ, 40^\circ, 50^\circ$ and 60° respectively. In each case make a drawing on a fresh page of drawing paper.

Questions

- (i) Record the value of θ , MN' , ON' , $\frac{MN'}{ON'}$ and $\cos\theta$
- (ii) Plot the graph of $\frac{MN'}{ON'}$ against $\cos\theta$
- (iii) Find the slope G of the graph
- (iv) Calculate the refractive index η given that $G = \frac{1}{\eta}$
- (v) State two sources of errors